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NOTE:
Before using this document please make sure that it is the latest revision.

This cover sheet controls the revision status of this entire document.

The revision of all sheets of this document will be at the same level and approvals shall be performed electronically.

Revision #	Name	Date	Description	PCN#
A	Resa Lambert	10Oct2003	Design Release	
B	Resa Lambert	14Nov2003	Added IG1 ping	
C	Resa Lambert	14Nov2003	Updates: 3-1, 4-1, 4-5, 4-10, 4-16	
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1				
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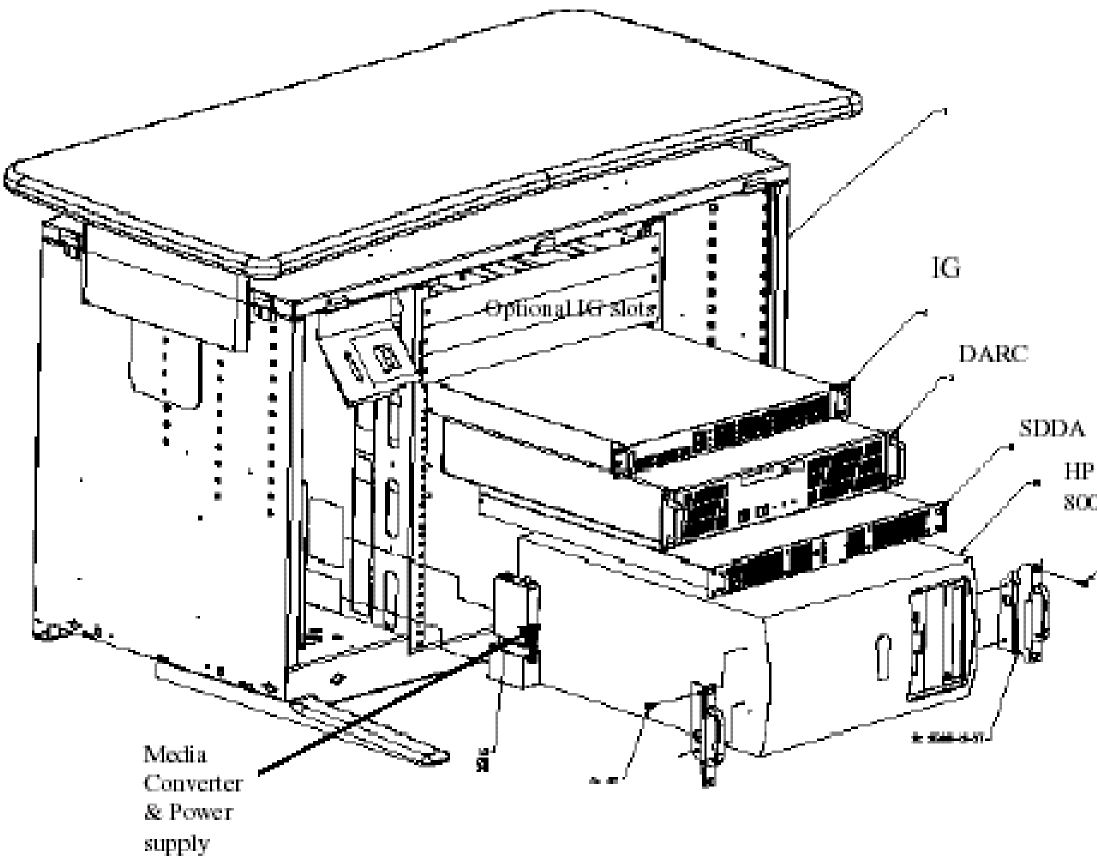
1. SCOPE

This engineering preliminary training procedure is used to perform various troubleshooting techniques on the various subsystems of the GRE system.



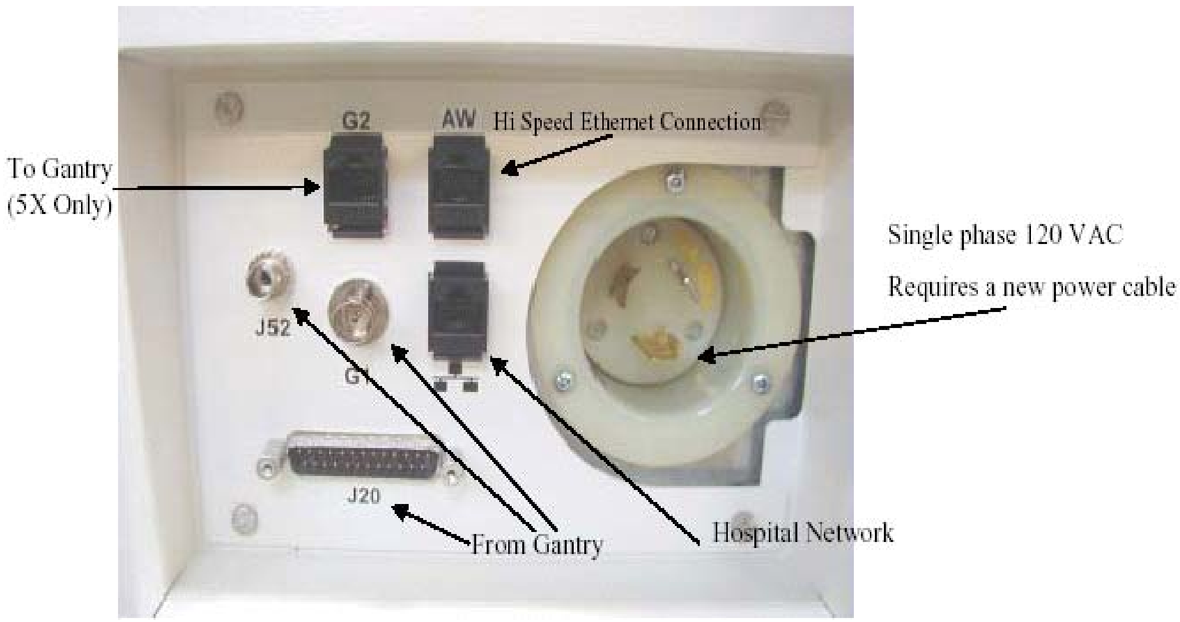
2. REQUIREMENTS

Verify and record specific system hardware configuration.

- Open a shell. Type: **cat /usr/g/config/INFO <Enter>**
  Record information on the GRE Information Checklist provided.

Verify the proper revision / type of Software has been loaded onto the GRE Console successfully or the DARC is having problems with the software load.

Ethernet Connections to the rear of the GRE Operator Console are installed and good.



3. GENERAL INFORMATION

GRE	Global Recon Engine
DARC	Data Acquisition and Recon Control Node
SDDA	Scan Data Disk Arra
IG1	Image Generator #1 Node —– the GRE Console is always shipped with #1
IG2	Image Generator #2 Node —– future (listed as fail during boot-up if unpopulated)
IG3	Image Generator #3 Node —– future (listed as fail during boot-up if unpopulated)
RAC	Recon Accelerator Card located in the IG Node
DIP	DAS Interface Processor located in the DARC Node —– specific to DAS type
SOL	Serial Over LAN
rsh	remote shell
halt	Console is taken fully down with the only option being power OFF.

3-1 GRE —— Information Checklist

Open a shell and type: **cat (space) /GEMS (then select Tab key several times)<Enter>**

Record the Operating System Version (1.7.9 or 1.7.10, etc): ______________

Verify and record specific system hardware configuration, this is a requirement for troubleshooting:

Open a shell and type: **cat (space) /usr/g/config/INFO <Enter>**

{ctuser@hostname}[1] cat /usr/g/config/INFO
setenv SERVICE_ID GE_SERVICE
setenv HOSPITAL_NAME ______________
setenv MENUTZ "America" ______________
setenv SUNTZ "US/Central" ______________
setenv TZ "CST6CDT US" ______________
setenv TZTEXT "Chicago" ______________
setenv REGENERATE_DATABASE no
setenv NEXT_MOD_NUMBER 1
setenv NEXT_DIAGNOSTIC_EXAM_NUMBER 50000
setenv GATEWAY_HOSTNAME ______________
setenv GATEWAY_**IP** ______________
setenv GATEWAY_**NETMASK** ______________
setenv GATEWAY_**BROADCAST** ______________
setenv **GATEWAY_DEFAULT** ______________
setenv **TUBETYPE** 15 / 16 / 17 ______________
setenv **PDUTYPE** CT_Compact / NG PDU ______________
setenv DASMTYPE 2
setenv KV_VALUES "80 100 120 140" ______________
setenv DEFAULTARCHIVE DICOM
setenv ADVANTAGERW FALSE
setenv CTIDATETIMEFORMAT 4
setenv TIMESEPAR /
setenv WEIGHT lb
setenv DOCTITLE "radiologist"
setenv CAMERATYPE ""
setenv SUITEID ______________
setenv FILTERTYPE 2
setenv CAMERA no
setenv DEFFORMAT 12
setenv DOGREYSCALE yes
setenv DOSLIDES no
setenv DOZOOM no
setenv FILMINTERP sharp

setenv PRINTER_HOSTNAME ________________
setenv PRINTER_IPADDRESS ________________
setenv PRINTER no
setenv O2_PRESENT n
setenv **GANTRY_TYPE** h2 / hp/ wb ________________
setenv LANGUAGE en_US
setenv DOSE y
setenv NETWORK yes
setenv USENIS yes
setenv CHGDARCSUBNET yes
setenv DOMAINNAME ________________
setenv NIS_SERVER ________________
setenv DARC_SUBNET ________________
setenv NEXT_PATIENT_EXAM_NUMBER 1
setenv MOBILESYSTEM no ________________
setenv STREAK y
setenv INROOMSCAN off
setenv AAA n
setenv AUTOMA_INDEX_TABLE_TYPE type2 ________________
setenv **RECONHARDWARE** GRE / No O2 Pegasus ________________
setenv **DASTYPE** SDAS or MDAS 4 / 8/ 16 ________________
setenv DETECTORTYPE Warp3
setenv REFRESH_RATE 76

setenv yp s
setenv ypmaster ${SUITEID}_OC0
setenv HOSTNAME SBC0
setenv ALIAS ${SUITEID}_SBC0
setenv UNSYSID $GATEWAY_HOSTNAME
setenv SERVID $GATEWAY_HOSTNAME
setenv tubeType $TUBETYPE
setenv ACRNEMA xxxx
setenv REGEN no

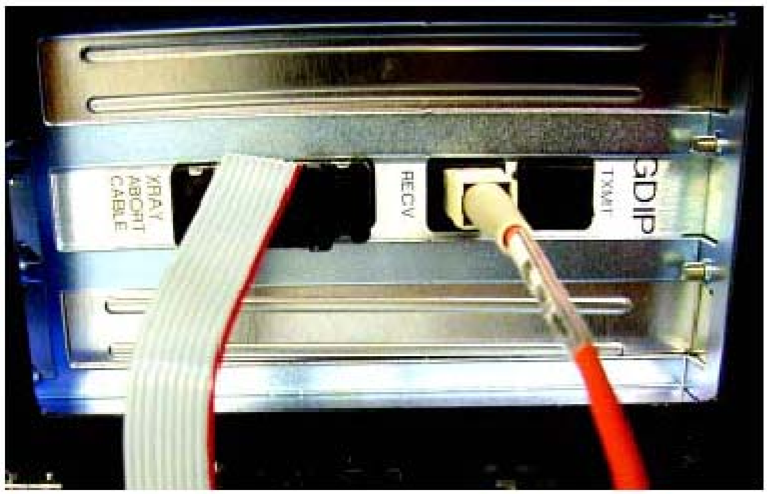
setenv HRCTTITLE ________________
setenv HRPORTNUM ________________
setenv HRAETITLE ________________
setenv HRIPADDR ________________
{ctuser@hosatname}

4. GRE Subsystem Testing

This engineering preliminary procedure is used to perform a series of tests or scripts to verify the DARC Node, IG Node and SDDA (plus some internal assemblies) are operating properly. Other tests may be available but were not outlined in this document. Most of the information seen in the Unix Shells has been copied from actual GRE Consoles and not input via the computer keyboard by a user.

4-1 DARC Node —— Cable Checkout

1. Verify the rear power cable is connected, switch is turned on and fan is operating.
2. Verify front power LED is illuminated.



3. Verify the Fiber Optic cable is connected to the DIP Board RX or RECV port (shown above – left most port) and it is in good condition.
4. Verify the 9-pin Sub D Scan XRAY Abort Cable is connected between the DIP Board and the SDDA (towards the right).
5. Verify the RHARD Interlock cable is connected to the 9-pin Sub D and directly below to the SDDA.
6. Verify the SCSI cable is connected to the SDDA and not kinked or in poor condition.

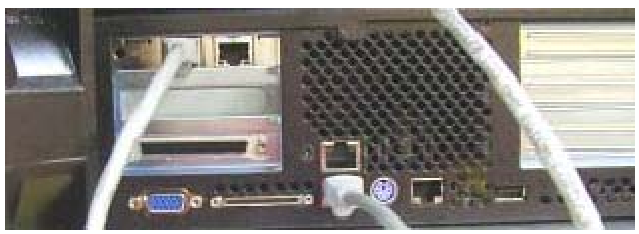
7. Verify Ethernet cable is in good condition and connected as follows:

- IG1 Ethernet 1 (bottom-left port) —— DARC Node Ethernet 0 (top-left port)

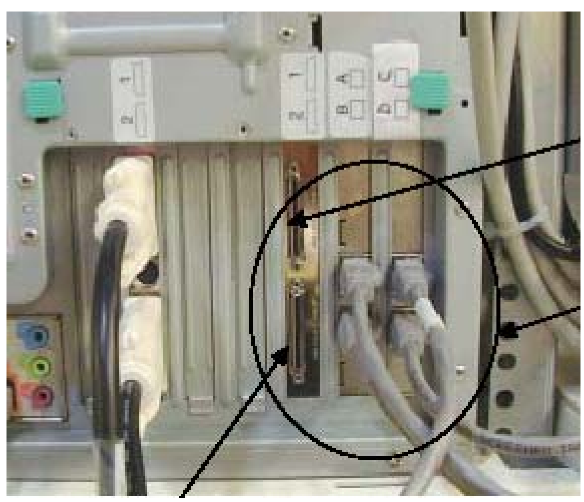
8. Verify Ethernet cable is in good condition and connected as follows:

- HP (right-card – top port) Ethernet C —— DARC Node Ethernet 3 (bottom-left port)
-

(DARC Node eth3 shown here – top connection)



(HP Host Computer Ethernet C shown here – see label)



4-2 DARC Node —– Ping and Remote Shell

This subsection will verify the DARC Node can be communicated too or pinged. After verifying the communication line exists between the HP Host PC and the DARC Node a remote shell will be invoked to determine if the DARC Node is up.

Open a Unix Shell and type the following:

- {ctuser@hostname} **su - <Enter>**
- Password: **#bigguy <Enter>**
- [root@hostname]# **ping darc <Enter>**
- *PING darc (172.16.0.2) from 172.16.0.1 : 56(84) bytes of data.*
- *64 bytes from darc (172.16.0.2): icmp_seq=1 ttl=64 time=0.157 ms*
- *64 bytes from darc (172.16.0.2): icmp_seq=2 ttl=64 time=0.209 ms*
- *64 bytes from darc (172.16.0.2): icmp_seq=3 ttl=64 time=0.175 ms*
- *64 bytes from darc (172.16.0.2): icmp_seq=4 ttl=64 time=0.134 ms*
- **Select: Control-C to stop the ping**
- *--- darc ping statistics ---*
- *4 packets transmitted, 4 received, 0% loss, time 3000ms*
- r*tt min/avg/max/mdev = 0.134/0.168/0.209/0.031 ms*

- [root@hostname]# **rsh darc <Enter>**
- *Last login: Thu Oct 9 11:17:35 on ttyS1*
- *You have new mail.*
- [root@darc ~]# **exit <Enter>**
- *logout*
- *rlogin: connection closed.*
- [root@hostname]#

4-3 DARC Node —— Serial Over LAN Checkout

Verification of the Serial Over LAN check is the almost always the initial test performed on the DARC after power and cable verification. It has been placed after ping and remote shell checkout because they are very quick tests. With SOL Checkout the telnet connection is made and the SOL connection from the HP Host Computer to the DARC Node is then established. This test will verify the DARC Node was booted and is up.

Open a Unix Shell and type the following:

{ctuser@hostname} **su - <Enter>**

Password: **#bigguy <Enter>**

[root@hostname]# **telnet localhost 623 <Enter>**

Trying 127.0.0.1...

Connected to localhost. ***** Indicates HP Host Apps were loaded *****

***** Indicates cliservice proxy server is running *****

Escape character is '^]'.

Server: **darc <Enter>**

Username: **<Enter>**

Password: **<Enter>**

Login successful ***** Indicates the SOL connection is established *****

dpccli> **console <Enter>**

GEMS Linux 1.7.9 ***** Console Operating System Software Revision *****

See you on the darc side of the moon. ***** DARC Node is up *****

darc login: **root <Enter>** ***** see note *****

Password: **#bigguy <Enter>**

Last login: Thu Oct 9 11:11:15 on ttyS1

You have new mail.

[root@darc ~]# **~.** **(tilde period to terminate telnet)**

dpccli> **exit <Enter>**

Connection closed by foreign host.

[root@hostname]#

NOTE

If the darc login prompt is not present and localhost.localdomain login prompt is displayed then try reloading the DARC Node OS and DARC Application software per the procedure. Problem could be the DARC Apps did not load well. Always be sure the DARC OS matches the HP Host OS already loaded.

4-4 DARC Node —– Telnet Connection Failed

If the telnet connection is fails this subsection will guide you to the next steps in identifying the root cause (s).

Open a Unix Shell and type the following:

{ctuser@hostname} **su - <Enter>**

Password: **#bigguy <Enter>**

[root@hostname]# **telnet localhost 623 <Enter>**

Trying 127.0.0.1...

Failed to connect to localhost.

[root@hostname]# **/etc/rc.d/init.d/cliservice <Enter>**

[root@hostname]# **telnet localhost 623 <Enter>**

Trying 127.0.0.1...

Connected to localhost.

Escape character is '^]'.

Server:

Discussion:

If the telnet connection is successful then perform SOL Checkout again.

If the telnet connection fails:

- Open a Unix Shell and halt the system
- Power Off the GRE Operator Console
- Wait 30 seconds.
- Power On the GRE Operator Console
- Retry the telnet command

If it still fails

- load the SOL CD if available for current version of Software.
- check or replace eth3 connection (DARC Node to HP Host ethA).
- reload the software package
- replace the HP Host PC or DARC Node as suggested by the OLC.

4-5 DARC Node —— Reset

The DARC Node reset –c test should only be performed if the dpccli> console command failed to return the *OS revision information* and *See you on the darc side of the moon* message. Whenever this test is performed the user must verify the information on the screen as it scrolls past looking for failures.

NOTE

During the DARC Node reboot or reset a [Failure] may appear which is acceptable. One type of acceptable failure would be 'Stopping NIS' which may fail because NIS was never running. Record all failures or errors and contact OLC for further assistance if necessary.

Open a Unix Shell and type the following:

- {ctuser@hostname} **su - <Enter>**
- Password: #**bigguy <Enter>**

[root@hostname]# **telnet localhost 623 <Enter>**

Trying 127.0.0.1...

Connected to localhost.

Escape character is '^]'.

Server: **darc <Enter>**

Username: **<Enter>**

Password: **<Enter>**

Login successful ***** Indicates the SOL connection is established *****

dpccli> **reset -c <Enter>**

The DARC Node will be reset. Below is a partial display screen displayed during the reset – this is loaded with **OS 1.7.9** version (each OS version will be different):

Linux Software loading

Linux version 2.4.18-24SGI_XFS_1.2.0smp (root@bob.ctt.med.ge.com) (gcc version 2.96 20000731 (Red Hat Linux 7.3 2.96-110)) #1 SMP Thu Feb 20 20:52:49 CST 2003

BIOS-provided physical RAM map:

BIOS-e820: 0000000000000000 - 000000000009b400 (usable)

BIOS-e820: 000000000009b400 - 00000000000a0000 (reserved)

BIOS-e820: 00000000000e8000 - 0000000000100000 (reserved)

BIuser: 0000000000000000 - 000000000009b400 (usable)

user: 000000000009b400 - 00000000000a0000 P was found.

ACPI: RSDP located at physical address c00ff9b0

RSD PTR v0 [INTEL]

__va_rangeIC (acpi_id[0x0000] id[0x0] enabled[1])

CPU 0 (0x0000) enabledProcessor #0 Pentium 4(tm) XEON(400] global_irq_base[0x30])INT_SRC_OVR (bus[0] irq[0x0] global_irq[0x2] polarity[0x0] triggerrocessor Specification v1.4 command: ~.

Virtual Wire compatibility mode.

OEM ID: INTEL   Product ID:5539.46 BogoMIPS

Memory: 1930604k/1966080k available (1694k kernel code, 30848k reserved, 1144k data, 188k init, 1048576k highmem)

Dentry cache hash table entries: 262144 (order: 9, 2097152 bytes)

Inode cache hash table entries: 131072 (order: 8, 1048576 bytes)

Mount cache hash ta done.

Checking 'hlt' instruction... OK.

POSIX conformance testing by UNIFIX

mtrr: v1.40 (20processor 1/1 eip 2000

Initializing CPU#1

masked ExtINT on CPU#1

ESR value before enabling v: Intel(R) Xeon(TM) CPU 2.80GHz stepping 07

Booting processor 2/6 eip 2000

Initializing CPU#25573.87 BogoMIPS

CPU: L1 I cache: 0K, L1 D cache: 8K

CPU: L2 cache: 512K

Setting 10 in the phys_id_present_map

...changing IO-APIC physical APIC ID to 10 ... ok.

..T..... CPU clock speed is 2790.1331 MHz.

..... host bus clock speed is 132.1737 MHz.

cpu: 0, c88,D:8,S:51906,C:259533>

CPU1<T0:259520,T1:155696,D:12,S:51906,C:259533>

checking TSC synchromigration_task 1 on cpu=1

PCI: PCI BIOS revision 2.10 entry at 0xfdb85, last bus=4

PCI: Using) -> 16

PCI->APIC IRQ transform: (B0,I29,P1) -> 19

PCI->APIC IRQ transform: (B0,I31,P0) -> 17isapnp: No Plug & Play device found

speakup: initialized device: /dev/synth, node (MAJOR 10, MINOR 25)

Linux NET4.0 for Linux 2.4

Based upon Swansea University Computer Society NET3.039

Initializing RT netlink socket

apm: BIOS not found.

Starting kswapd

allocated 64 pages and 64 bhs reserved for the highmem bounces

VFS: Disk quotas vdquot_6.5.1

SGI XFS 1.2.0 with ACLs Unix98 ptys configured

Serial driver version 5.05c (2001-07-08) with MANY_PORTS MULTIPORT SHARE_IRQ SERIAL_PCI ISAPNP enabled

ttyS0 at 0x03f8 (irq = 4) is a 16550A

ttyS1 at 0x02f8 (irq = 3) is a 16550A

Real Time Clock Driver v1.10e

oprofile: can't get RTC I/O Ports

block: 1024 slots per queue, batch=256

Uniform Multi-Platform E-IDE driver Revision: 6.31

ide: Assuming 33MHz system bus speed for PIO modes; overrides later

ide0: BM-DMA at 0x03a0-0x03a7, BIOS settings: hda:DMA, hdb:DMA

ide1: BM-DMA ahda: ST340014A, ATA DISK drive

hdb: SR244W, ATAPI CD/DVD-ROM drive

ide0 at 0x1f0-0x1f7,0x3f6 on irq 14

blk: queue c048ea44, I/O limit 4095Mb (mask 0xffffffff)

blk: queue c048ea44, I/O limit 4095Mb (mask 0xffffffff)

hda: 78165360 sectors (40021 MB) w/2048KiB Cache, CHS=4865/255/63, UDMA(100)

ide-floppy driver 0.99.newide

Partition check:

hda: hda1 hda2 hda3 hda4 < hda5 hda6 >

Floppy drive(s): fd0 is 1.44M

FDC 0 is a National Semiconductor PC87306

NET4: Frame Diverter 0.46

RAMDISK driver initialized: 16 RAM disks of 4096K size 1024 blocksize

ide-floppy driver 0.99.newideXFS mounting filesystem ide0(3,2)

Starting XFS recovery on filesystem: ide0(3,2) (dev: 3/2)

Ending XFS recovery on filesystem: ide0(3,2) (dev: 3/2)

VFS: Mounted root (xfs filesystem) readonly.

Freeing unused kernel memory: 188k freed

INIT: version 2.84 booting

Welcome to GEMS Linux

Press 'I' to enter interactive startup.

Mounting proc filesystem: [OK]

Configuring kernel parameters: [OK]

```
Setting clock  (localtime): Thu Oct  9 11:07:11 CDT 2003 [  OK  ]

Activating swap partitions:  [  OK  ]

Setting hostname darc:  [  OK  ]

Mounting USB filesystem:  [  OK  ]

Initializing USB controller (usb-uhci):  [  OK  ]

Your system appears to have shut down uncleanly

Press Y within 1 seconds to force file system integrity check...

Checking root filesystem

[/sbin/fsck.xfs (1) -- /] fsck.xfs -a /dev/hda2

[  OK  ]

Remounting root filesystem in read-write mode:  [  OK  ]

Finding module dependencies:  [  OK  ]

Checking filesystems

Checking all file systems.

[/sbin/fsck.xfs (1) -- /boot] fsck.xfs -a /dev/hda1

[/sbin/fsck.xfs (1) -- /usr/g] fsck.xfs -a /dev/hda6

[  OK  ]

Mounting local filesystems:  [  OK  ]

Enabling local filesystem quotas:  [  OK  ]

Enabling swap space:  [  OK  ]

INIT: Entering runlevel: 3

Entering non-interactive startup

Setting network parameters:  [  OK  ]

Bringing up loopback interface:  [  OK  ]

Bringing up interface eth0:  [  OK  ]
```

Bringing up interface eth1: [OK]

Bringing up interface eth2: [OK]

Bringing up interface eth3: [OK]

Starting system logger: [OK]

Starting kernel logger: [OK]

Starting portmapper: [OK]

Starting NFS statd: [OK]

Loading system font: [OK]

Initializing random number generator: [OK]

Mounting NFS filesystems: [OK]

Mounting other filesystems: [OK]

Starting dip: dip: found dip type 0x3229 i=2

dip: memtop   =0x78000000

dip: high_memory =0xf8000000

dip: __pa(highmem)=0x38000000

dip: DMA bus address 0x78000000 ; high mem at 0x38000000

dip: DIP base addr 0xe0000000 ioremap addr 0xf8b42000 reg_base=0x8000000

dip: Detected MDAS 16-Slice DIP: 2326523 B

[OK]

Starting igpower1: [OK]

Starting igpower2: Starting ig2:[FAILED]

[FAILED]

Starting igpower3: Starting ig3:[FAILED]

[FAILED]

Starting xinetd: [OK]

Starting NFS services: [OK]

Starting NFS quotas: [OK]

Starting NFS mountd: [OK]

Starting NFS daemon: [OK]

Starting bmcscript: [OK]

Starting dhcpd: [OK]

Starting console mouse services: [OK]

Starting crond: [OK]

Starting anacron: [OK]

Starting atd: [OK]

Starting cliservice: [OK]

GEMS Linux 1.7.9 ***** Console Operating System Software Revision *****
See you on the darc side of the moon. ***** DARC Node is up *****
darc login: **root <Enter>**
Password: **#bigguy <Enter>**
Last login: Thu Oct 9 11:05:08 on ttyS1
You have new mail.
root@darc ~]# **~.** **(tilde period to terminate telnet)**

dpccli> **exit <Enter>**
Connection closed by foreign host.
[root@hostname]#

4-6 DARC Node —— ifconfig eth3 Plus

This subsection will confirm the address at eth2 and the DARC Node Subnet address. Specific sites may require the DARC Subnet to be changed due to a conflict with the 172 base address which is initially loaded onto the HP Host and DARC Node form the Load From Cold software installation process.

Open a Unix Shell and type the following:

- {ctuser@hostname} **ifconfig <Enter>**

eth2 inet addr: 169 254.0.2 Bcast: 169.254.0.255 Mast: 255.255.255.0

- {ctuser@hostname} **su - <Enter>**
- Password: **#bigguy <Enter>**
- [root@hostname]# **rsh darc <Enter>**
- *Last login: Thu Oct 9 11:17:35 on ttyS1*
- *You have new mail.*
- [root@darc ~]# **ifconfig eth3 <Enter>**
- inet addr: 172 16.0.2 Bcast: 172.16.0.255 Mast: 255.255.255.0

NOTE

The eth3 inet addr and Bcast may be different for specific sites that have experienced a 172 conflict. These specific sites will have a different address. One example is shown below at the end.

- [root@darc ~]# **exit <Enter>**
- [root@hostname]#

DARC Subnet Address Example:

[root@darc ~]# **ifconfig eth3 <Enter>**

inet addr: 169 254.0.2 Bcast: 169.254.0.255 Mast: 255.255.255.0

4-7 DARC Node —— DIP Diagnostic Help

The DIP Board is located in the DARC Node. There are two connections to the DIP Board. A Fiber Optic Cable connects to RX or RECV port (left-most connection as viewed from the rear of the GRE Console). A 9-pin SubD Scan XRAY Abort Cable is also present which connects to the SDDA.

DIP Diagnostics are run via the DARC Node. A help menu is available as shown below. DIP Diagnostics may be run individually or as a group of all. Additionally, a number can be entered for iterations (DipDiag –a –i 5 <Enter>).

Test # 7 and test #8 will fail without a loopback cable installed between RX/RECV and TX/TXMT.

Loopback Cable installation: The Loopback cable should be known good before use in troubleshooting. Disconnect the Fiber Optic cable coming from the Gantry at the rear Console bulkhead connector. Connect one end of the loopback cable to the the rear Console bulkhead connector. Connect the other end of the loopback cable to the TX port of the DARC Node DIP Board. If Tests 7 and 8 fail – bypass the internal Console cable by connecting the loopback cable between the RX and TX ports of the DIP Board if possible.

Open a Unix Shell and type the following:

{ctuser@hostname} **rsh darc <Enter>**

Last login: Tue Oct 7 14:58:34 from oc

[ctuser@darc ~]$ **cd /usr/g/bin <Enter>**

[ctuser@darc bin]$ **./DipDiag –h <Enter>**

usage: DipDiag [options]

options:

-a, --all,
-x, --execute, 1,2,3,...
-n, --do-not-execute, 1,2,3,...
-i, --iterations, N
-q, --quiet,
-p, --stop-on-error,
-h, --help,

Test numbers are as follows

1 - X-ray Enable Relay
2 - Test FEC Bit
3 - Test EDDR Bit
4 - Test Magic Number Register
5 - Memory Test
6 - Interrupt Test
7 - External LoopBack Test
8 - DMA Test

Without options the test is run in interactive mode.

[ctuser@darc bin]$

4-8 DARC Node —— DIP Diagnostic All

The DIP Board is located in the DARC Node. There are two connections to the DIP Board. A Fiber Optic Cable connects to RX or RECV port (left-most connection as viewed from the rear of the GRE Console). A 9-pin SubD Scan XRAY Abort Cable is also present which connects to the SDDA.

DIP Diagnostics are run via the DARC Node. A help menu is available. DIP Diagnostics may be run individually or as a group of all. Additionally, a number can be entered for iterations (DipDiag –a –i 5 <Enter>).

Test # 7 and test #8 will fail without the use of a loopback cable.

{ctuser@hostname} **rsh darc <Enter>**

Last login: Tue Oct 7 14:58:34 from oc

[ctuser@darc ~]$ **cd /usr/g/bin <Enter>**

[ctuser@darc bin]$ **./DipDiag –a <Enter>**

12:05:40 opening LINUX device /dev/dip

creating H16 dip interface

DIP PN : 2326523 B *****Specific DIP Board identified here *****

test 1 Passed X-ray Enable Relay

test 2 Passed Test FEC Bit

test 3 Passed Test EDDR Bit

test 4 Passed Test Magic Number Register

fill mem

write to dip took 4.714 sec

verifying memory

read from dip took 19.469 sec

test 5 Passed Memory Test

test 6 Passed Interrupt Test

error - failed memcmp of pattern

test 7 Failed External LoopBack Test

error - DIP detected an error 0x69

test 8 Failed DMA Test

1 loops 2 failures ***** See NOTE *****

2 tests failed

NOTE

Test #7 and test #8 will fail without a loopback cable installed between RX and TX of the DIP Board.

4-9 DARC Node —— DIP Diagnostic Individual Testing

The DIP Board is located in the DARC Node. There are two connections to the DIP Board. A Fiber Optic Cable connects to RX or RECV port (left-most connection as viewed from the rear of the GRE Console). A 9-pin SubD Scan XRAY Abort Cable is also present which connects to the SDDA.

DIP Diagnostics are run via the DARC Node. A help menu is available. DIP Diagnostics may be run individually. Additionally, a number can be entered for iterations (DipDiag –6 –i 5 <Enter>).

Test # 7 and test #8 will fail without the use of a loopback cable.

In the test case shown below only tests 1 through 6 were selected because a loopback cable was not installed on the Console. If a loopback cable is available select 12345678.

Open a Unix Shell and type the following:

{ctuser@hostname} **rsh darc <Enter>**

Last login: Tue Oct 7 14:58:34 from oc

[ctuser@darc ~]$ **cd /usr/g/bin <Enter>**

[ctuser@darc bin]$ **./DipDiag <Enter>**

```
+---------------------------------------------------+
|                                                   |
|                     WARNING                       |
|                    =========                      |
|                                                   |
| - THIS TOOL IS FOR MANUFACTURING PROCESSES ONLY.  |
| - DO NOT USE THIS TOOL WHILE SCANNING.            |
| - APPLICATION PROCESSES SHOULD BE SHUTDOWN WHILE  |
|   USING THIS TOOL.                                |
+---------------------------------------------------+
```

Do you want to continue? (Y/N): **y <Enter>**

12:05:40 opening LINUX device /dev/dip

creating H16 dip interface

info - ***'DIP Board Existence Test' was successful***

info - ***DIP board revision is '2326523-1B'***

```
+----------------------------+
| DIP Diagnostics Main Menu |
+----------------------------+
```

1. X-ray Enable Relay

2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: **123456 <Enter>** ***** No loopback cable is installed *****
+info - ***'X-ray Enable Test' was successful***

```
+---------------------------+
| DIP Diagnostics Main Menu |
+---------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: info - ***'FEC Bit Test' was successful***

```
+---------------------------+
| DIP Diagnostics Main Menu |
+---------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test

7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: info - ***'EDDR Bit Test' was successful***

```
+---------------------------+
| DIP Diagnostics Main Menu |
+---------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: info - ***'Magic Number Register Test' was successful***

```
+---------------------------+
| DIP Diagnostics Main Menu |
+---------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: fill mem
write to dip took 4.714 sec

verifying memory
read from dip took 19.469 sec
info - ***'Memory Test' was successful***

```
+--------------------------+
| DIP Diagnostics Main Menu |
+--------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: info - ***'Interrupt Test' was successful***

```
+--------------------------+
| DIP Diagnostics Main Menu |
+--------------------------+
```

1. X-ray Enable Relay
2. Test FEC Bit
3. Test EDDR Bit
4. Test Magic Number Register
5. Memory Test
6. Interrupt Test
7. External LoopBack Test
8. DMA Test
0. Exit

Please make your selection: **0 <Enter>**

Exiting DIP Diagnostics.

[ctuser@darc bin]$ **exit <Enter>**
logout
rlogin: connection closed.
{ctuser@hostname}

4-10 IG Node —— Cable Checkout

1. Verify the rear power cable is connected, switch is turned on and fan is operating.
2. Verify front power LED is illuminated.
3. Verify Ethernet cabling is not kinked and in good condition and connected as follows:

- IG1Ethernet 1 (bottom-left port) —— DARC Node Ethernet 0 (top-left port)

4-11 IG Node —— Ping and Remote Shell

This subsection will verify the IG Node can be communicated too or pinged via the DARC Node. After verifying the communication line exists between the IG Node and the DARC Node a remote shell will be invoked to determine if the IG Node is up. This can be performed as ctuser or as root.

Open a Unix Shell and type the following:

- {ctuser@hostname} **su - <Enter>**
- Password: **#bigguy <Enter>**
- [root@hostname]# **rsh darc <Enter>**
  Last login: Thu Oct 9 11:17:35 from oc
  You have new mail.
- [root@darc ~]# **ping ig1 <Enter>**
- *PING ig1 (10.0.1.2) from (10.0.1.1): 56(84) bytes of data.*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=1 ttl=64 time=0.157 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=2 ttl=64 time=0.209 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=3 ttl=64 time=0.175 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=4 ttl=64 time=0.134 ms*
- **Select: Control-C to stop the ping**
- *--- ig1 ping statistics ---*
- *4 packets transmitted, 4 received, 0% loss, time 3000ms*
- r*tt min/avg/max/mdev = 0.134/0.168/0.209/0.031 ms*

[root@darc ~]# **rsh ig1 <Enter>**
Password: **#bigguy <Enter>**
Last login: Thu Oct 16 13:04:55 from darc

- [root@ig1 ~]# **ping ig1 <Enter>**
- *PING ig1 (10.0.1.1) from (10.0.1.2): 56(84) bytes of data.*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=1 ttl=64 time=0.157 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=2 ttl=64 time=0.209 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=3 ttl=64 time=0.175 ms*
- *64 bytes from ig1 (10.0.1.2) icmp_seq=4 ttl=64 time=0.134 ms*
- **Select: Control-C to stop the ping**

- *--- ig1 ping statistics ---*
- *4 packets transmitted, 4 received, 0% loss, time 3000ms*
- r*tt min/avg/max/mdev = 0.134/0.168/0.209/0.031 ms*
- [root@ig1 ~]# **exit <Enter>**
- *logout*
- *rlogin: connection closed.*
- [root@darc ~]# **exit <Enter>**

[root@ig1 ~]# **exit <Enter>**

- *logout*
- *rlogin: connection closed.*
- [root@hostname]#

4-12 IG Node —— SOL connection from DARC Node to IG1

The DARC Node controls the IG Node. The DARC Node must be up.

Open a Unix Shell and type the following:
{ctuser@hostname} **su - <Enter>**
Password: **#bigguy <Enter>**
[root@hostname]# **rsh darc<Enter>**
[root@hostname]# **telnet localhost 623 <Enter>**
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
Server: **ig1 <Enter>**
Username: **<Enter>**
Password: **<Enter>**
Login successful ***** Indicates the SOL connection from DARC to IG1 is established *****

dpccli> **exit <Enter>**

4-13 IG Node —— Remote Shell to IG1 and RAC Board Information

The RAC Board resides in the IG Node, however, the DARC Node controls the IG Node. The DARC Node must be up to view the flash revision (fan_para-6-12) and the device driver information.

Open a Unix Shell and type the following:

{ctuser@hostname} **su - <Enter>**

Password: **#bigguy <Enter>**

[root@hostname]# **rsh darc<Enter>**

Last login: Wed Oct 8 11:20:22 from oc

[ctuser@darc ~]$ **ls /usr/local/gemsrac <Enter>**

fan_para-6-12.hexout power_up_cfg-6-11.hexout

****** fan_para-6-12.hexout is the filename we will use to see the actual information ******

[ctuser@darc ~]$ **rsh ig1 <Enter>**

Last login: Wed Oct 8 11:20:51 from darc

For convienence - here is theFLASH help menu:

```
usage [options] flash_rac hexfile

options:
-d device   Specify the device name (default
/dev/gemsrac)
-e          Erase FLASH before programming.
-n          Do not write to the FLASH.
-f          Force programming, do not check the image
first.
-v          Verify the FLASH against the hex input file.
            and do nothing else.
-l n        Write, erase, and/or verify FPGA load n
-o filename Save binary output to file.
-S n        Program only the first n sectors (for debug)
```

[ctuser@ig1 ~]$ **flash_rac –v –l 1 *fan_para-6-12.hexout* <Enter>**

```
Board ID: 2316826 C
done reading file 'fan_para-6-12.hexout'
1305558 bytes
Calculated checksum byte is 0x7e
image size is padded to 1310720
verify OK
```

[ctuser@ig1 ~]$ **cat /proc/driver/gemsrac <Enter>**

RAC Device Driver Version 1.2

compiled Oct 3 2003 05:14:04

Board Physical Identifier: 2316826 C

PCI Command Reg : 0x117

PCI Error Status: 0x0
PCI Rev   : **0xe**   or (0x8 with the 0x0 below)
PCI Subsys   : **0x1**   or (0x0 with the 0x8 above)
DMA bus address : 0x78000000
[ctuser@ig1 ~]$ **exit <Enter>**
logout
rlogin: connection closed.
[ctuser@darc ~]$ **exit <Enter>**
logout
rlogin: connection closed.
{ctuser@hostname}

4-14 IG Node —– RAC Board Diagnostics

The RAC Board resides in the IG Node, however, the DARC Node controls the IG Node. The DARC Node must be up to view the flash revision (fan_para-6-12) and the device driver information.

NOTE

RAC diagnostic Test 3 SRAM B Incr Pat will always fail the first pass run and must pass subsequent passes.

Open a Unix Shell and type the following:
{ctuser@hostname} **cd /usr/g/bin <Enter>**
{ctuser@hostname} **ls rac* <Enter>**
racdiags*
{ctuser@hostname} **racdiags <Enter>**
9 (counts from 0 to 9)
 0 SRAM A Incr Pat   PASSED 10 loops
 1 SRAM A Walking Pat   PASSED 10 loops
 2 SRAM A Address Pat   PASSED 10 loops
 3 SRAM B Incr Pat   PASSED 10 loops   ***** This will fail on the initial pass ONLY *****
 4 SRAM B Walking Pat   PASSED 10 loops
 5 SRAM B Address Pat   PASSED 10 loops
 6 FLASH Region 0   PASSED 10 loops
 7 FLASH Region 1   PASSED 10 loops
13 DMA host<-image mem   PASSED 10 loops
14 DMA host->projection mem   PASSED 10 loops

10 Loops: All tests passed
{ctuser@hostname}

4-15 IG Node —— racbist Diagnostics

The racbist diagnostics are available in command line form. These are the same as racdiags except that the racdiags have been pre-set to run a series of 2 tests initially and not report their fail status. The tests that may initially may fail are # 0 and # 3. For illustrative purposes we will show an actual test file output which shows the initial failure of test #3. If racbist diagnostics are to be utilized it is recommended that test #0 and test #3 are run initially and then further diagnostics are run.

Also, racdiags do not run the complete list of racbist tests.

Open a Unix Shell and type the following:

{ctuser@hostname} **su - <Enter>**

Password: **#bigguy <Enter>**

[root@hostname]# **rsh darc <Enter>**

Last login: Thu Oct 16 13:40:38 from oc

You have new mail.

[root@darc ~]# **rsh ig1 <Enter>**

Password: **#bigguy <Enter>**

Last login: Thu Oct 16 13:04:55 from darc

[root@ig1 ~]# **racbist <Enter>**

0 SRAM A Incr Pat	**PASSED 1 loops**
1 SRAM A Walking Pat	PASSED 1 loops
2 SRAM A Address Pat	PASSED 1 loops
3 SRAM B Incr Pat	**FAILED 1 of 1**
4 SRAM B Walking Pat	PASSED 1 loops
5 SRAM B Address Pat	PASSED 1 loops
6 FLASH Region 0	PASSED 1 loops
7 FLASH Region 1	PASSED 1 loops
8 FLASH Region 2	PASSED 1 loops
9 FLASH Region 3	PASSED 1 loops
10 FLASH Region 4	PASSED 1 loops
11 FLASH Region 5	PASSED 1 loops
12 DMA host->image mem	PASSED 1 loops
13 DMA host<-image mem	PASSED 1 loops
14 DMA host->projection mem	PASSED 1 loops
15 DMA host<-projection mem	PASSED 1 loops

```
[root@ig1 ~]#  racbist --ex 0,3 <Enter>                (run this initially as these may fail)
  0 SRAM A Incr Pat           PASSED 1 loops
  3 SRAM B Incr Pat           PASSED 1 loops
1 Loops: All tests passed

[root@ig1 ~]# racbist --ex 0,1,2,3,4,5,6,7,13,14 --it 10 <Enter>
9
  0 SRAM A Incr Pat           PASSED 10 loops
  1 SRAM A Walking Pat        PASSED 10 loops
  2 SRAM A Address Pat        PASSED 10 loops
  3 SRAM B Incr Pat           PASSED 10 loops
  4 SRAM B Walking Pat        PASSED 10 loops
  5 SRAM B Address Pat        PASSED 10 loops
  6 FLASH Region 0            PASSED 10 loops
  7 FLASH Region 1            PASSED 10 loops
 13 DMA host<-image mem       PASSED 10 loops
 14 DMA host->projection mem  PASSED 10 loops

10 Loops: All tests passed
[root@ig1 ~]# exit <Enter>
logout
rlogin: connection closed.
[root@darc ~]# exit <Enter>
logout
rlogin: connection closed.
[root@hostname]#
```

4-16 SDDA —— Cable Checkout

1. Verify the rear power cable is connected, switch is turned on and fan is operating.
2. Verify front power LED is illuminated.
3. Verify the 9-pin Sub D Scan Abort cable is connected between the DIP Board and the SDDA (towards the right). Verify DIP errors are present during reboot when application software is started).
4. Verify the Interlock cable is connected to the 9-pin Sub D and directly above to the DARC Node.
5. Verify the SCSI cable is connected to the DARC Node and not kinked or in poor condition,(verify no SCSI - RAID errors are present during reboot when application software is started)
6. Verify the two Audio connections to the HP Host Computer are good.

4-17 SDDA —— View Error Message LOG

Visually scan the Error Message Log for Scan Data Disk failures. Record the Error message, time, and error number for reference during the troubleshooting process.

4-18 DARC Node —— SDDA Utilities

The DARC Node controls the SDDA. The DARC Node must be up.

Open a Unix Shell and type the following:

{ctuser@hostname} **rsh (space) darc <Enter>**

Last login: Tue Oct 7 14:58:34 from oc

[ctuser@darc ~]$ **cd (space) /usr/g/scripts <Enter>**

{ctuser@darc scripts} **./SDDAUtils(space) -U <Enter>**

could be a lot of already mounted info – look for the OK below:

[Logged by SDDAUtils] Mounted /dev/md0 at /raw_data. [OK]

{ctuser@darc scripts} **cat(space) /proc/scsi/scsi <Enter>**

```
Attached devices:
Host: scsi0 Channel: 00 Id: 00 Lun: 00
  Vendor: MITSUMI  Model: CD-ROM SR244W    Rev: T01A
  Type:   CD-ROM                           ANSI SCSI revision: 02
Host: scsi1 Channel: 00 Id: 00 Lun: 00
  Vendor: SEAGATE  Model: ST336753LW       Rev: 0005
  Type:   Direct-Access                    ANSI SCSI revision: 03
Host: scsi1 Channel: 00 Id: 01 Lun: 00
  Vendor: SEAGATE  Model: ST336753LW       Rev: 0005
  Type:   Direct-Access                    ANSI SCSI revision: 03
```

{ctuser@darc scripts} **exit <Enter>**

4-19 SDDA —– testHighSpeed

The DARC Node mounts the SDDA drives. The SDDA houses two Scan Data Disk Drives and the Intercom / Prescribed Tilt Board. This test verifies the SCSI cabling between the DARC Node and the SDDA. It writes data to the drive, then reads is back and verifies a checksum looking for dropped packets. It ensures the disk is able to transfer data at a rate equal to or greater than 75 MB/sec. This test shall be run with Applications down.

Visually verify the front panel SDDA LEDs for HD1 and HD2 blink during the test.

Open a Unix Shell and type the following:

{ctuser@hostname} **rsh darc <Enter>**

Last login: Tue Oct 7 15:12:51 from oc

[ctuser@darc ~]$ **cd /usr/g/bin <Enter>**

[ctuser@darc bin]$ **ls test* <Enter>**

testHighSpeed

[ctuser@darc bin]$ **./testHighSpeed <Enter>**

Starting pass_number 1
saving 1024 MB to disk
1024 MB in 10.480 sec 97.710 MB/sec

Starting pass_number 2
saving 1024 MB to disk
1024 MB in 10.030 sec 102.094 MB/sec

Starting pass_number 3
saving 1024 MB to disk
1024 MB in 9.340 sec 109.636 MB/sec

Starting pass_number 4
saving 1024 MB to disk
1024 MB in 8.150 sec 125.644 MB/sec

Starting pass_number 5
saving 1024 MB to disk
1024 MB in 8.440 sec 121.327 MB/sec

Average speed: 111.282 Mb/sec. testHighSpeed PASSED.

[ctuser@darc bin]$ **exit <Enter>**
logout
rlogin: connection closed.
{ctuser@hostname}

4-20 SDDA —— reconfigScanDisk

The DARC Node mounts the SDDA drives. The SDDA houses two Scan Data Disk Drives and the Intercom / Prescribed Tilt Board. This test wipes out all Scan Data and reconfigures the Scan Disk. This test shall be run with Applications down.

Open a Unix Shell and type the following:

{ctuser@hostname} **su - <Enter>**

Password: #**bigguy <Enter>**

[root@hostname]#]# **rsh darc <Enter>**

Last login: Tue Oct 7 14:42:40 from oc

You have new mail.

[root@darc ~]# **cd /usr/g/scripts <Enter>**

[root@darc scripts]# **ls reconfig* <Enter>**

reconfigScanDisk.log *reconfigScanDisk*

[root@darc scripts]# **./reconfigScanDisk <Enter>**

Appears that /raw_data is mounted.Trying to unmount..OK

Checking if SCSI driver [aic79xx] is loaded..

SCSI driver is already loaded. Good.

Checking if primary disk is IDE..

Detected IDE disk as primary.Good.

Detecting other SCSI disks on the system..

Detected 2 SCSI disk(s) : /dev/sda /dev/sdb

Creating primary partition on /dev/sda

Creating primary partition on /dev/sdb

Creating entries in /etc/raidtab.gre..OK

Initializing RAID device..OK

Examining if I created /dev/md0 correctly..OK

Creating XFS filesystem..OK

Creating mount point directory : /raw_data ..OK

Verifying what I did..

Mounting RAID disks at /raw_data..MountDevice

/dev/md0 on /raw_data type xfs (rw,nodev)

Verifying mount , Status = 1

[should be 1]

[Logged by SDDAUtils] Mounted /dev/md0 at /raw_data. [OK]

OK

Creating a file /raw_data/test and writing to it..OK

Deleting file /raw_data/test..OK

Verification completed successfully

Changing ownership of /raw_data to ctuser [owner] and ctuser [group]..OK
Applying permissions [777] to /raw_data..OK
All done. Transcript saved in reconfigScanDisk.log
[root@darc scripts]# **df <Enter>**
Filesystem 1k-blocks Used Available Use% Mounted on
/dev/hda2 4188164 1238768 2949396 30% /
/dev/hda1 27296 11488 15808 43% /boot
none 967508 16412 951096 2% /dev/shm
/dev/hda6 30647188 40080 30607108 1% /usr/g
oc:/usr/g/queue/prospective
14935616 2052336 12883280 14% /usr/g/queue/prospective
oc:/usr/g 14935616 2052336 12883280 14% /usr/g/oc
/dev/md0 71673044 2605820 69067224 4% /raw_data

NOTE * If /raw_data is not present then perform the ./startSDDA and df again*****

[root@darc scripts]# **./startSDDA <Enter>**
[root@darc scripts]# **df <Enter>**
Filesystem 1k-blocks Used Available Use% Mounted on
/dev/hda2 4188164 1238768 2949396 30% /
/dev/hda1 27296 11488 15808 43% /boot
none 967508 16412 951096 2% /dev/shm
/dev/hda6 30647188 40080 30607108 1% /usr/g
oc:/usr/g/queue/prospective
14935616 2052336 12883280 14% /usr/g/queue/prospective
oc:/usr/g 14935616 2052336 12883280 14% /usr/g/oc
/dev/md0 71673044 2605820 69067224 4% ***/raw_data***
[root@darc scripts]# **exit <Enter>**
rlogin: connection closed.
[root@bayhostname]#